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JAVA PROJECT REPORT ON

# COMPLAINT MANAGEMENT AND RESOLUTION TRACKING SYSTEM

Submitted by

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# DECLARATION

I hereby declare that the project entitled "VEHICLE SERVICE REMAINDER AND MAINTAINENCE TRACKER" submitted in partial fulfillment of the requirements for the course JAVA PROGRAMMING is a **record of my original work** carried out under the guidance of **DHILEEPAN A**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

RITHIKA SRI R

# CERTIFICATE

This is to certify that the project entitled **"VEHICLE SERVICE REMAINDER AND MAINTAINENCE TRACKER"** is a **bona fide work** carried out by RITHIKA SRI R**, I st YEAR**, , B.Sc IT - A, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

**Place:**

**Date:**

Submitted for the Viva-Voce Examination held on _____

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## TABLE OF CONTENTS


| S. No. | Topics | Page. No. |
|---|---|---|
| **1** | **Introduction** | |
| 1.1 | Objective | |
| 1.2 | Overview | |
| 1.3 | Features | |
| **2** | **Program Analysis** | |
| 2.1 | Problem Definition | |
| 2.2 | Project Overview | |
| 2.3 | Scopes and Limitations | |
| **3** | **System Design** | |
| 3.1 | Modules | |
| **4** | **Methodology** | |
| 4.1 | Problem Analysis | |
| 4.2 | System Design | |
| 4.3 | Program Development | |
| 4.4 | Testing and Validation | |
| 4.5 | Result and Documentation | |
| 4.6 | Algorithm | |
| 4.7 | Flowchart | |
| **5** | **Source Code** | |
| **6** | **Input and Output Screens** | |
| **7** | **Discussion and Conclusion** | |
| 7.1 | Discussion | |
| 7.2 | Conclusion | |
| **8** | **References** | |

# ABSTRACT

The Vehicle Service Reminder and Maintenance Tracker is a Java-based application developed to help vehicle owners track their service schedules efficiently. Many vehicle users forget their vehicle service dates or kilometer limits, which can lead to poor vehicle performance and increased maintenance costs.

This project provides a simple solution where users can enter vehicle details such as vehicle name, owner name, last service kilometer reading, and current kilometer reading. Based on these inputs, the system calculates the next service due kilometer automatically and reminds the user whether servicing is required or not.

The system is developed using Core Java concepts such as classes, objects, methods, Scanner class, and conditional statements. This project demonstrates the practical implementation of Java programming in solving real-world maintenance tracking problems.

# 1. INTRODUCTION

In today's busy lifestyle, many vehicle owners forget their vehicle servicing schedules due to lack of proper tracking methods. Regular servicing is important for maintaining vehicle performance and increasing its lifespan.

The Vehicle Service Reminder and Maintenance Tracker is designed to automate the service reminder process. It helps users store vehicle details and check whether servicing is required based on kilometer readings.

This system is developed using Core Java and works as a console-based application. It improves maintenance efficiency and reduces manual tracking effort.

## 1.1 Objective

The main objectives of this project are:

- To develop a vehicle service reminder system using Java
- To accept vehicle details from users
- To store last service kilometer reading
- To accept current kilometer reading
- To calculate next service kilometer automatically
- To display reminder message when service is due
- To reduce manual tracking effort
- To demonstrate implementation of Core Java concepts

## 1.2 Overview

The Vehicle Service Reminder and Maintenance Tracker is a simple console-based application.

The system accepts:

- Vehicle name
- Owner name
- Last service kilometer
- Current kilometer

Then the system calculates:

**Next Service Kilometer = Last Service Kilometer + 3000**

After comparison, the system displays whether servicing is required.

This application helps vehicle owners maintain proper service schedules easily.

## 1.3 Features

The main features include:

• Accept vehicle details
• Store service kilometer reading
• Automatic calculation of next service kilometer
• Service reminder notification
• Easy-to-use console interface
• Helps prevent missing service schedules

# 2. PROGRAM ANALYSIS

Program analysis is an important stage in software development. It helps understand the problem and design an efficient solution.

This system solves the issue of missing vehicle service schedules by providing an automated reminder system.

### 2.1 Problem Definition

Manual complaint management systems face several issues:

- Complaints may be lost or not recorded properly
- No proper tracking of complaint status
- Delay in resolving complaints
- Difficulty in managing large number of complaints

To overcome these issues, a computerized system is required.

### 2.2 Project Overview

The Vehicle Service Reminder System is designed to help users track service schedules easily.

Functions of the system include:

• Accept vehicle details
• Store last service kilometer
• Accept current kilometer
• Calculate next service kilometer
• Display reminder message

The system improves maintenance tracking efficiency.

### 2.3 Scopes and Limitations

## SCOPE:

- This system can be used in:

• Personal vehicle tracking
• Bike maintenance tracking
• Car service tracking
• Small service centers

It helps users maintain vehicle service schedules properly.

## LIMITATIONS:

- Tracks only one vehicle at a time
  • No database connectivity
  • No service history storage
  • Console-based interface only

Future improvements can include database and GUI features.

# 3. SYSTEM DESIGN

System design describes the structure of the application before implementation.

The system is divided into modules for better organization.

### 3.1 Modules

The main modules are explained below:

**1. Input Module**

This module collects:

• Vehicle name
• Owner name
• Last service kilometer
• Current kilometer

Scanner class is used to receive input.

**2. Calculation Module**

This module calculates:

Next Service Kilometer = Last Service Kilometer + 3000

**3. Reminder Module**

This module checks whether service is required.

Condition used:

If current kilometer ≥ next service kilometer
Display reminder message

Otherwise
Display service not required message

**4. Display Module**

This module displays:

• Vehicle details
• Next service kilometer
• Reminder message

# 4. METHODOLOGY

Methodology explains steps followed during system development.

Steps include:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result Documentation

## 4.1 Problem Analysis

Manual service tracking methods are not reliable.

Problems include:

• Forgetting service schedules
• No reminder system
• Difficulty maintaining records

This system solves these problems using automation.

## 4.2 System Design

The system consists of four modules:

• Input Module
• Calculation Module
• Reminder Module
• Display Module

These modules work together to perform service tracking operations efficiently.

## 4.3 Program Development

The system was developed using Core Java concepts such as:

• Classes
• Objects
• Methods
• Scanner class
• Conditional statements

The program compares kilometer readings and generates reminder messages automatically.

## 4.4 Testing and Validation

Testing ensures the system works correctly.

Example test cases:

| Test Case | Expected Result |
|---|---|
| Enter vehicle name | Stored correctly |
| Enter last service km | Stored correctly |
| Enter current km | Compared correctly |

| Test Case | Expected Result |
|---|---|
| Service limit crossed | Reminder displayed |

The system produced correct outputs during testing.

## 4.5 Result and Documentation

After execution, the system displays:

• Vehicle name
• Owner name
• Next service kilometer
• Reminder status

The system works correctly for different inputs.

Documentation includes:

• System design
• Source code
• Input/output results

## 4.6 Algorithm

1. Step-by-step procedure:

Start program
Enter vehicle name
Enter owner name
Enter last service kilometer
Enter current kilometer
Calculate next service kilometer
Compare with current kilometer

Display reminder message

End program

## 4.7 Flowchart

Draw this in your record:

Start

↓

Enter vehicle details

↓

Calculate next service km

↓

Compare with current km

↓

Display reminder message

↓

End

# 5. SOURCE CODE

```
import java.util.Scanner;

class Vehicle
{
    String vehicleName;
    String ownerName;
    int lastServiceKm;
    int currentKm;
```

```
void getDetails()
{
    Scanner sc = new Scanner(System.in);

    System.out.println("Enter Vehicle Name:");
    vehicleName = sc.nextLine();

    System.out.println("Enter Owner Name:");
    ownerName = sc.nextLine();

    System.out.println("Enter Last Service Kilometer:");
    lastServiceKm = sc.nextInt();

    System.out.println("Enter Current Kilometer:");
    currentKm = sc.nextInt();
}

void checkService()
{
    int nextService = lastServiceKm + 3000;

    System.out.println("\nVehicle Name: " + vehicleName);
    System.out.println("Owner Name: " + ownerName);
    System.out.println("Next Service Due at: " + nextService + " km");

    if(currentKm >= nextService)
    {
        System.out.println("Service Reminder: Your vehicle needs servicing now!");
```

```
        }
        else
        {
            System.out.println("Service Status: No service needed now.");
        }
    }
}

public class VehicleServiceTracker
{
    public static void main(String args[])
    {
        Vehicle v = new Vehicle();

        System.out.println("Vehicle Service Reminder System\n");

        v.getDetails();
        v.checkService();
    }
}
```

## 6. INPUT AND OUTPUT SCREENS

# 7.DISCUSSION & CONCLUSION

### 7.1 Discussion

The Vehicle Service Reminder and Maintenance Tracker was successfully developed using Core Java programming concepts.

The system helps users track service schedules easily and reduces manual effort.

Testing confirmed that the system produces correct results.

### 7.2 Conclusion

The Vehicle Service Reminder and Maintenance Tracker is a simple and effective application developed using Java.

It helps vehicle owners maintain service schedules properly and prevents missing maintenance intervals.

Future improvements may include:

• Database connectivity
• Multiple vehicle tracking
• GUI interface

# 8. REFERENCES

The following resources were used during the development of this project:

**Books**


- *E. Balagurusamy – Programming with Java, McGraw-Hilly*
- *Y. Daniel Liang – Introduction to Java Programming, Pearson*

**Websites**

- https://www.geeksforgeeks.org
- https://www.w3schools.com/java